Hanif Carroll

Applied AI Engineer | LLM systems, evaluations & product engineering

U.S. citizen · Buenos Aires, Argentina · U.S. hours · Open to remote employment or contracts through HC Studio LLC

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Summary

Applied AI Engineer with eight years of full-stack software engineering experience. Builds production LLM systems — hybrid retrieval, evaluations, agent tool use via MCP, and durable AI workflows — and owns the product around them. Has shipped for early-stage startups, an established SaaS platform, and enterprise clients including Disney, Cisco, and GE.

Technical Skills

Applied AI: Anthropic & OpenAI APIs, RAG / hybrid retrieval & reranking, pgvector, structured outputs, tool calling / MCP, LLM & multimodal evaluations, LangGraph, LangSmith, Amazon Bedrock

Backend & Languages: Python, FastAPI, Pydantic, TypeScript, Node.js, PostgreSQL, Supabase, Zod

Cloud & Infrastructure: AWS (Step Functions, SQS, DynamoDB), Cloudflare Workers / Workflows, Vercel, Docker, GitHub Actions

Frontend & Mobile: React, Next.js, TanStack Start, SvelteKit, React Native / Expo, Playwright

Work Experience

Founder & AI Product Engineer **HC Studio** **Mar 2024 – Present**

- Built and operated independent AI products and delivered client products with founders, covering product design, software engineering and operations.
- Partnered directly with founders to launch Palabruno on web, iOS and Android and help Genrupt serve its first 200 paying customers.

Casamo — AI Product Engineer

- Built an AI accommodation-research product that turns listings, reviews and photos into cited shortlists, with evaluations scoring exact source quotations, visual false positives and private-versus-shared-space classification.
- Built durable listing research on Cloudflare Workflows, with checkpoints that preserved completed work and kept queue delays from triggering premature timeouts.

BA Eventos — AI Product Engineer

- Built conversational event search for web and Telegram across a six-source catalog, combining semantic and keyword retrieval with venue and date filters.
- Built regression evaluations to check that recommendations stayed grounded in the event catalog and follow-up questions referred to the right events.

Palabruno — AI Product Engineer

- Built a Spanish text-processing pipeline combining linguistic rules and constrained LLM generation to create contextual learning cards, with output validation, targeted regeneration, regression tests, and per-stage latency and model-cost tracking.

- Built a multi-stage story generator that structures teacher constraints, plans scenes, and validates and repairs vocabulary, grammar, length and narrative defects.

Genrupt — AI Product Engineer

- Built recoverable AI generation workflows with atomic PostgreSQL credit reservations and row locking, reusing existing provider jobs and preventing duplicate credit settlement.
- Enabled agents to use seller workflows through MCP, with OAuth authentication, organization-level access controls and asynchronous results.

Mucho Hangouts — Fractional CTO

- Served as Fractional CTO on a five-person team, mentoring two junior developers through code reviews, pair programming and reusable frontend patterns.

Senior Frontend Engineer **LeagueApps, later via Nearsure** **Nov 2021 – Feb 2024**

- Modernized tournament scheduling with React on the company's first migration team, helping establish shared frontend patterns and components.
- Built drag-and-drop scheduling, conflict detection, and double-booking prevention, reducing organizer scheduling time by 45%.
- Built self-service reporting with AG Grid; used virtualization and memoization for tens of thousands of rows, reducing load and interaction lag by 70%.

Software Engineer **Capgemini** **Jan 2019 – Oct 2021**

- Built reporting and KPI dashboards for Disney Cruise Line and Cisco; maintained Java Spring Boot microservices for GE Aviation.

Frontend Engineer **8base** **Nov 2017 – Nov 2018**

- Built Jira / GitLab CI/CD automation and refactored 20+ shared UI components into a reusable library, reducing duplicate code by 30%.

Selected Projects

Switchboard — AI change-management demo | Live demo

- Built a Python/FastAPI and LangGraph agent that gathers permission-scoped evidence and checks policy interpretations before proposing changes for human approval.
- Deployed the demo on AWS using Step Functions, SQS and DynamoDB, with transactional configuration updates and signed delivery verification.

Languages

- **English** — Native
- **Spanish** — Professional working proficiency

Education

- **B.S. in Health Sciences,** University of South Florida